

LEANN with Ollama: RAG Everything on Your Laptop: Easy Tutorial



Fahd Mirza
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This video is a step-by-step Tutorial to install and use LEANN with Ollama, which is a vector database for laptop.

← → ↺

https://github.com/yichuan-w/LEANN

150%

README Contributing MIT license



LEANN

Python 3.9 | 3.10 | 3.11 | 3.12 | 3.13 CI passing Platform Ubuntu & Arch & WSL | macOS (ARM64/Intel) License MIT

MCP Native Integration Slack Join WeChat Join

The smallest vector index in the world. RAG Everything with LEANN!

LEANN is an innovative vector database that democratizes personal AI. Transform your laptop into a powerful RAG system that can index and search through millions of documents while using **97% less storage** than traditional solutions **without accuracy loss**.

LEANN achieves this through *graph-based selective recomputation with high-degree preserving pruning*, computing embeddings on-demand instead of storing them all. [Illustration Fig →](#) | [Paper →](#)

Ready to RAG Everything? Transform your laptop into a personal AI assistant that can semantic search your [file system](#), [emails](#), [browser history](#), [chat history](#), [codebase*](#), or external knowledge bases (i.e., 60M documents) - all on your laptop, with zero cloud costs and complete privacy.

← → ↺

https://github.com/yichuan-w/LEANN

🔍 Type to search

🔖 +

Code Issues 25 Pull requests 2 Discussions Actions Projects Security Insights

LEANN (Public)

Watch 24 Fork 254 Star 2.7k

main 46 Branches 27 Tags

Go to file Add file Code

andylzf

Introducing dynamic index update (#108)

5f7806e · 6 hours ago

391 Commits

.github	feat: Add GitHub PR and issue templates (#105)	3 days ago
.vscode	Auto-register apps: Universal index discovery (#64)	last month
apps	[Fix] Enable AST chunking when installed (package chunking...	4 days ago
assets	[chore] add slack to share use case	3 weeks ago
benchmarks	[EXP] Update the benchmark code (#71)	last month
data	fix missing file	last month
docs	Ast fork (#92)	2 weeks ago
examples	Introducing dynamic index update (#108)	6 hours ago
packages	Introducing dynamic index update (#108)	6 hours ago
scripts	fix ruff errors and formatting	2 months ago
sky	Docs/Core: Low-Resource Setups, SkyPilot Option, and No-R...	last month
tests	Metadata filtering feature (#75)	last month
videos	add gif	2 months ago

RAG on Everything with LEANN. Enjoy 97% storage savings while running a fast, accurate, and 100% private RAG application on your personal device.

python privacy ai offline-first localstorage vectors faiss rag vector-search vector-database llamain langchain llama-index retrieval-augmented-generation ollama gpt-oss

Readme MIT license Contributing Activity 2.7k stars 24 watching 254 forks Report repository Releases 26

```
Ubuntu@0360-kci2-ty6k-prxmx100037: ~  
Ubuntu@0360-kci2-ty6k-prxmx100037:~$ ollama list  
NAME                                ID                SIZE    MODIFIED  
embeddinggemma:latest              85462619ee72     621 MB   33 minutes ago  
gpt-oss:20b                         aa4295ac10c3     13 GB    21 hours ago  
Ubuntu@0360-kci2-ty6k-prxmx100037:~$
```

```
Ubuntu@0360-kci2-ty6k-prxmx100037:~$ conda create -n ai python=3.11 -y && conda activate ai  
Channels:  
- defaults  
Platform: linux-64  
Collecting package metadata (repodata.json): \
```

```
#  
# $ conda deactivate  
  
(ai) Ubuntu@0360-kci2-ty6k-prxmx100037:~$ conda install -c conda-forge --override-channels notebook -y  
conda install -c conda-forge --override-channels ipywidgets -y  
  
pip install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu121  
pip install git+https://github.com/huggingface/transformers  
pip install git+https://github.com/huggingface/accelerate  
pip install huggingface_hub  
Channels:  
- conda-forge  
Platform: linux-64  
Collecting package metadata (repodata.json): \
```

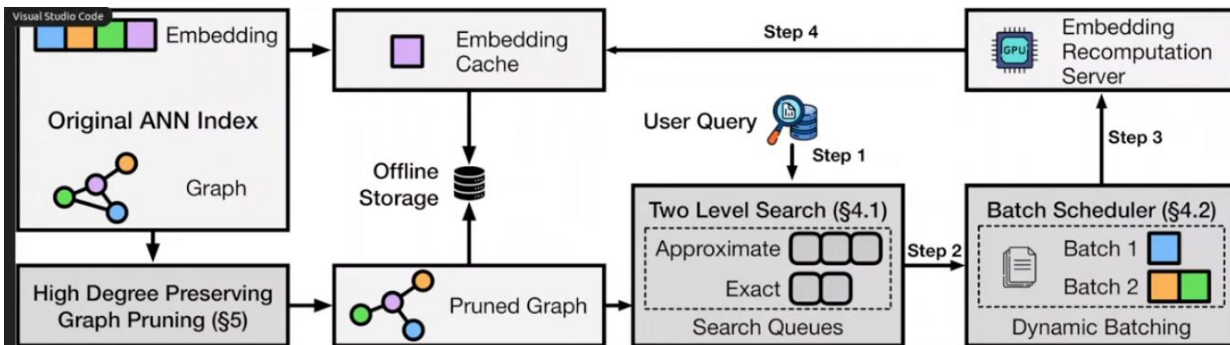
```
Ubuntu@0360-kci2-ty6k-prxmx100037: ~/leann  
(ai) Ubuntu@0360-kci2-ty6k-prxmx100037:~$ git clone https://github.com/yichuan-w/LEANN.git leann && cd leann  
Cloning into 'leann'...  
remote: Enumerating objects: 6741, done.  
remote: Counting objects: 100% (695/695), done.  
remote: Compressing objects: 100% (179/179), done.  
remote: Total 6741 (delta 588), reused 529 (delta 513), pack-reused 6046 (from 3)  
Receiving objects: 100% (6741/6741), 67.78 MiB | 42.42 MiB/s, done.  
Resolving deltas: 100% (3633/3633), done.  
(ai) Ubuntu@0360-kci2-ty6k-prxmx100037:~/leann$
```

```
Ubuntu@0360-kci2-ty6k-prxmx100037: ~/leann  
(ai) Ubuntu@0360-kci2-ty6k-prxmx100037:~/leann$ uv --version  
uv 0.8.15  
(ai) Ubuntu@0360-kci2-ty6k-prxmx100037:~/leann$
```

```
Ubuntu@0360-kci2-ty6k-prxmx100037: ~/leann  
(ai) Ubuntu@0360-kci2-ty6k-prxmx100037:~/leann$ uv venv  
Using CPython 3.11.13  
Creating virtual environment at: .venv  
Activate with: source .venv/bin/activate  
(ai) Ubuntu@0360-kci2-ty6k-prxmx100037:~/leann$ source .venv/bin/activate  
(leann) (ai) Ubuntu@0360-kci2-ty6k-prxmx100037:~/leann$ uv pip install leann  
.i nvidia-cufile-cu12==1.13.1.3
```

```
Ubuntu@0360-kci2-ty6k-prxmx100037: ~/leann  
(leann) (ai) Ubuntu@0360-kci2-ty6k-prxmx100037:~/leann$ uv pip install ollama  
Resolved 13 packages in 127ms  
Prepared 1 package in 22ms  
Installed 1 package in 1ms  
+ ollama==0.5.3  
(leann) (ai) Ubuntu@0360-kci2-ty6k-prxmx100037:~/leann$
```

```
Ubuntu@0360-kci2-ty6k-prxmx100037: ~/leann  
(leann) (ai) Ubuntu@0360-kci2-ty6k-prxmx100037:~/leann$ jupyter notebook  
[I 2025-09-13 22:11:34.953 ServerApp] jupyter_lsp | extension was successfully linked.  
[I 2025-09-13 22:11:34.956 ServerApp] jupyter_server_terminals | extension was successfully linked.  
[I 2025-09-13 22:11:34.959 ServerApp] jupyterlab | extension was successfully linked.  
[I 2025-09-13 22:11:34.962 ServerApp] notebook | extension was successfully linked.
```



```

[1]: from leann import LeannBuilder, LeannSearcher, LeannChat
from pathlib import Path

INDEX_PATH = str(Path(".").resolve() / "demo.leann")

[ ]: # === STEP 1: BUILD INDEX WITH embeddinggemma:latest ===
builder = LeannBuilder(
    backend_name="hnsf",
    embedding_model="embeddinggemma:latest", # Exact model name from ollama list
    embedding_mode="ollama",
    graph_degree=32,
    build_complexity=64
)
builder.add_text("LEANN saves 97% storage compared to traditional vector databases.")
builder.add_text("Tung Tung Tung Sahur called-they need their banana-crocodile hybrid back")
builder.build_index(INDEX_PATH)
print("Index built with embeddinggemma:latest (ollama)")

# === STEP 2: SEARCH - ALWAYS EXPECT A LIST ===
searcher = LeannSearcher(
    INDEX_PATH,
    embedding_model="embeddinggemma:latest", # MUST MATCH BUILD MODEL
    embedding_mode="ollama",
    search_complexity=32
)

```

```

# === STEP 2: SEARCH - ALWAYS EXPECT A LIST ===
searcher = LeannSearcher(
    INDEX_PATH,
    embedding_model="embeddinggemma:latest", # MUST MATCH BUILD MODEL
    embedding_mode="ollama",
    search_complexity=32
)
results = searcher.search("fantastical AI-generated creatures", top_k=1)

# FIXED: results is a list - access [0].text
print("Search result:", results[0].text if results else "No match")

# === STEP 3: CHAT WITH gpt-oss:20b ===
chat = LeannChat(
    INDEX_PATH,
    llm_config={"type": "ollama", "model": "gpt-oss:20b"},
    embedding_model="embeddinggemma:latest", # Match again
    embedding_mode="ollama",
    thinking_budget="medium"
)
response = chat.ask("How much storage does LEANN save?", top_k=1)
print("Answer:", response)

```



```
jupyter Untitled4 Last Checkpoint: 7 minutes ago
File Edit View Run Kernel Settings Help
JupyterLab Python 3 (ipykernel)

[0.30s] Compact key is false, reading original format...
[0.30s] Probing for potential extra byte before non-compact offsets...
[0.31s] Found and consumed an unexpected 0x00 byte.
      Reading vector (dtype=<class 'numpy.uint64'>, fmt='Q')... Count=3, Bytes=24
[0.31s] Read offsets (3)
[0.39s] Attempting to read neighbors vector...
      Reading vector (dtype=<class 'numpy.int32'>, fmt='i')... Count=128, Bytes=512
[0.39s] Read neighbors (128)
[0.48s] Read scalar params (ep=1, max_lvl=0)
[0.48s] Checking for storage data...
[0.48s] Found storage fourcc: 49467849.
[0.48s] Converting to CSR format...
[0.48s] Conversion loop finished.
[0.48s] Running validation checks...
      Checking total valid neighbor count...
      OK: Total valid neighbors = 2
      Checking final pointer indices...
      OK: Final pointers match data size.
[0.48s] Deleting original neighbors and offsets arrays.
```

```
jupyter Untitled4 Last Checkpoint: 7 minutes ago
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JupyterLab Python 3 (ipykernel)

      OK: Final pointers match data size.
[0.48s] Deleting original neighbors and offsets arrays...
      CSR Stats: |data|=2, |level_ptr|=4
[0.56s] Writing CSR HNSW graph data in FAISS-compatible order...
      Pruning embeddings: Writing NULL storage marker.

INFO:leann_backend_hnsw.hnsw_backend: CSR conversion successful.
[0.66s] Conversion complete.

INFO:leann_backend_hnsw.hnsw_backend:INFO: Replaced original index with CSR-pruned version at '/home/Ubuntu/leann/demo.index'
INFO:leann.embedding_server_manager:Terminating server process (PID: 166840) for backend leann_backend_hnsw.hnsw_embedding_server...

Index built with embeddinggemma:latest (Ollama)

INFO:leann.embedding_server_manager:Server process 166840 terminated gracefully.
INFO:leann.embedding_server_manager:Server process 166840 cleanup completed
INFO:leann.api: LeannSearcher.search() called:
INFO:leann.api: Query: 'fantastical AI-generated creatures'
INFO:leann.api: Top_k: 1
INFO:leann.api: Metadata filters: None
```

```
jupyter Untitled4 Last Checkpoint: 7 minutes ago
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JupyterLab Python 3 (ipykernel)

Index built with embeddinggemma:latest (Ollama)

INFO:leann.embedding_server_manager:Server process 166840 terminated gracefully.
INFO:leann.embedding_server_manager:Server process 166840 cleanup completed
INFO:leann.api: LeannSearcher.search() called:
INFO:leann.api: Query: 'fantastical AI-generated creatures'
INFO:leann.api: Top_k: 1
INFO:leann.api: Metadata filters: None
INFO:leann.api: Additional kwargs: {}
INFO:leann.embedding_server_manager:Starting embedding server on port 5557...
INFO:leann.embedding_server_manager:Command: /home/Ubuntu/leann/.venv/bin/python3 -m leann_backend_hnsw.hnsw_embedding_server --zmq-port 5557 --model-name embeddinggemma:latest --passages-file /home/Ubuntu/leann/demo.leann.meta.json --embedding-mode ollama --distance-metric mips
INFO:leann.embedding_server_manager:Starting server process with command: /home/Ubuntu/leann/.venv/bin/python3 -m leann_backend_hnsw.hnsw_embedding_server --zmq-port 5557 --model-name embeddinggemma:latest --passages-file /home/Ubuntu/leann/demo.leann.meta.json --embedding-mode ollama --distance-metric mips
INFO:leann.embedding_server_manager:Server process started with PID: 167046
INFO:leann.embedding_server_manager:Embedding server is ready!
INFO:leann.api: Launching server time: 1.5039381980895996 seconds
```

```
jupyter Untitled4 Last Checkpoint: 7 minutes ago
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JupyterLab Python 3 (ipykernel)

7 --model-name embeddinggemma:latest --passages-file /home/Ubuntu/leann/demo.leann.meta.json --embedding-mode ollama --distance-metric mips
INFO:leann.embedding_server_manager:Starting server process with command: /home/Ubuntu/leann/.venv/bin/python3 -m leann_backend_hnsw.hnsw_em
bedding_server --zmq-port 5557 --model-name embeddinggemma:latest --passages-file /home/Ubuntu/leann/demo.leann.meta.json --embedding-mode o
llama --distance-metric mips
INFO:leann.embedding_server_manager:Server process started with PID: 167046
INFO:leann.embedding_server_manager:Embedding server is ready!
INFO:leann.api: Launching server time: 1.5039381980895996 seconds
INFO:leann.embedding_server_manager:Reusing in-process server
INFO:leann.api: Generated embedding shape: (1, 768)
INFO:leann.api: Embedding time: 0.24895286560058594 seconds
INFO:leann_backend_hnsw.hnsw_backend: Search time in HNSWSearcher.search() backend: 0.391643762588501 seconds
INFO:leann.api: Search time in search() LEANN searcher: 0.39246034622192383 seconds
INFO:leann.api: Backend returned: labels=1 results
INFO:leann.api: Processing 1 passage IDs:
INFO:leann.api: ✓ [1] ID: '1' Score: 0.7553 Text: Tung Tung Tung Sahur called-they need their banana-crocodile hybrid back
INFO:leann.api: ✓ Final enriched results: 1 passages
INFO:leann.chat:Attempting to create LLM of type='ollama' with model='gpt-oss:20b'
INFO:leann.chat:Initializing OllamaChat with model='gpt-oss:20b' and host='http://localhost:11434'
```

```
jupyter Untitled4 Last Checkpoint: 7 minutes ago
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JupyterLab Python 3 (ipykernel)

[read_HNSW - CSR NL v4] Reading metadata & CSR indices (manual offset)...
[read_HNSW NL v4] Read levels vector, size: 2
[read_HNSW NL v4] Reading Compact Storage format indices...
[read_HNSW NL v4] Read compact_level_ptr, size: 4
[read_HNSW NL v4] Read compact_node_offsets, size: 3
[read_HNSW NL v4] Read entry_point: 1, max level: 0
[read_HNSW NL v4] Read storage fourcc: 0x6C6C756e
[read_HNSW NL v4 FIX] Detected FileIOReader. Neighbors size field offset: 242
[read_HNSW NL v4] Reading neighbors data into memory.
[read_HNSW NL v4] Read neighbors data, size: 2
[read_HNSW NL v4] Finished reading metadata and CSR indices.
INFO: Skipping external storage loading, since is_recompute is true.
ZmqDistanceComputer initialized: d=768, metric=0
M: 64 for level: 0
[read_HNSW - CSR NL v4] Reading metadata & CSR indices (manual offset)...
[read_HNSW NL v4] Read levels vector, size: 2
[read_HNSW NL v4] Reading Compact Storage format indices...
[read_HNSW NL v4] Read compact_level_ptr, size: 4
```

```
jupyter Untitled4 Last Checkpoint: 8 minutes ago
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JupyterLab Python 3 (ipykernel)

[read_HNSW NL v4] Read entry_point: 1, max level: 0
[read_HNSW NL v4] Read storage fourcc: 0x6C6C756e
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[read_HNSW NL v4] Read compact_level_ptr, size: 4
[read_HNSW NL v4] Read compact_node_offsets, size: 3
[read_HNSW NL v4] Read entry_point: 1, max level: 0
[read_HNSW NL v4] Read storage fourcc: 0x6C6C756e
[read_HNSW NL v4 FIX] Detected FileIOReader. Neighbors size field offset: 242
[read_HNSW NL v4] Reading neighbors data into memory.
[read_HNSW NL v4] Read neighbors data, size: 2
```

```
jupyter Untitled4 Last Checkpoint: 8 minutes ago
File Edit View Run Kernel Settings Help
JupyterLab Python 3 (ipykernel)

[read_HNSW - CSR NL v4] Reading metadata & CSR indices (manual offset)...
[read_HNSW NL v4] Read levels vector, size: 2
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[read_HNSW NL v4] Read compact_level_ptr, size: 4
[read_HNSW NL v4] Read compact_node_offsets, size: 3
[read_HNSW NL v4] Read entry_point: 1, max_level: 0
[read_HNSW NL v4] Read storage fourcc: 0x6c6c756e
[read_HNSW NL v4 FIX] Detected FileIOReader. Neighbors size field offset: 242
[read_HNSW NL v4] Reading neighbors data into memory.
[read_HNSW NL v4] Read neighbors data, size: 2
[read_HNSW NL v4] Finished reading metadata and CSR indices.
INFO: Skipping external storage loading, since is_recompute is true.
ZmqDistanceComputer initialized: d=768, metric=0
[read_HNSW - CSR NL v4] Reading metadata & CSR indices (manual offset)...
[read_HNSW NL v4] Read levels vector, size: 2
[read_HNSW NL v4] Reading Compact Storage format indices...
[read_HNSW NL v4] Read compact_level_ptr, size: 4
[read_HNSW NL v4] Read compact_node_offsets, size: 3
```

```
jupyter Untitled4 Last Checkpoint: 8 minutes ago
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JupyterLab Python 3 (ipykernel)

[read_HNSW NL v4] Read compact_level_ptr, size: 4
[read_HNSW NL v4] Read compact_node_offsets, size: 3
[read_HNSW NL v4] Read entry_point: 1, max_level: 0
[read_HNSW NL v4] Read storage fourcc: 0x6c6c756e
[read_HNSW NL v4 FIX] Detected FileIOReader. Neighbors size field offset: 242
[read_HNSW NL v4] Reading neighbors data into memory.
[read_HNSW NL v4] Read neighbors data, size: 2
[read_HNSW NL v4] Finished reading metadata and CSR indices.
INFO: Skipping external storage loading, since is_recompute is true.
ZmqDistanceComputer initialized: d=768, metric=0
M: 64 for level: 0
[read_HNSW - CSR NL v4] Reading metadata & CSR indices (manual offset)...
[read_HNSW NL v4] Read levels vector, size: 2
[read_HNSW NL v4] Reading Compact Storage format indices...
[read_HNSW NL v4] Read compact_level_ptr, size: 4
[read_HNSW NL v4] Read compact_node_offsets, size: 3
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[read_HNSW NL v4] Read storage fourcc: 0x6c6c756e
[read_HNSW NL v4 FIX] Detected FileIOReader. Neighbors size field offset: 242
```

```
jupyter Untitled4 Last Checkpoint: 8 minutes ago
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JupyterLab Python 3 (ipykernel)

[read_HNSW NL v4] Read entry_point: 1, max_level: 0
[read_HNSW NL v4] Read storage fourcc: 0x6c6c756e
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ZmqDistanceComputer initialized: d=768, metric=0
[read_HNSW - CSR NL v4] Reading metadata & CSR indices (manual offset)...
[read_HNSW NL v4] Read levels vector, size: 2
[read_HNSW NL v4] Reading Compact Storage format indices...
[read_HNSW NL v4] Read compact_level_ptr, size: 4
[read_HNSW NL v4] Read compact_node_offsets, size: 3
[read_HNSW NL v4] Read

INFO:leann.embedding_server_manager:Server process 166864 terminated gracefully.
INFO:leann.embedding_server_manager:Server process 166864 cleanup completed
INFO:leann.api:LeannSearcher.search() called:
INFO:leann.api:Query: Max num storage data (LEANN) 1000000
```



```
jupyter Untitled4 Last Checkpoint: 8 minutes ago
File Edit View Run Kernel Settings Help
JupyterLab Python 3 (ipykernel)
print(response)
INFO:leann.embedding_server_manager:Server process 166864 cleanup completed
INFO:leann.api: LeannSearcher.search() called:
INFO:leann.api: Query: 'How much storage does LEANN save?'
INFO:leann.api: Top_k: 1
INFO:leann.api: Metadata filters: None
INFO:leann.api: Additional kwargs: {}
INFO:leann.embedding_server_manager:Starting embedding server on port 5558...
INFO:leann.embedding_server_manager:Command: /home/Ubuntu/leann/.venv/bin/python3 -m leann_backend_hnsw.hnsw_embedding_server --zmq-port 5558 --model-name embeddinggemma:latest --passages-file /home/Ubuntu/leann/demo.leann.meta.json --embedding-mode ollama --distance-metric mips
INFO:leann.embedding_server_manager:Starting server process with command: /home/Ubuntu/leann/.venv/bin/python3 -m leann_backend_hnsw.hnsw_embedding_server --zmq-port 5558 --model-name embeddinggemma:latest --passages-file /home/Ubuntu/leann/demo.leann.meta.json --embedding-mode ollama --distance-metric mips
INFO:leann.embedding_server_manager:Server process started with PID: 167069
INFO:leann.embedding_server_manager:Embedding server is ready!
INFO:leann.api: Launching server time: 1.5045411586761475 seconds
INFO:leann.embedding_server_manager:Reusing in-process server
INFO:leann.api: Generated embedding shape: (1, 768)
INFO:leann.api: Embedding time: 0.2468271255493164 seconds
INFO:leann_backend_hnsw.hnsw_backend: Search time in HNWSearcher.search() backend: 0.3752765655517578 seconds
```

The screenshot shows a JupyterLab window titled "Untitled4" with a last checkpoint of 8 minutes ago. The interface includes a top bar with "File", "Edit", "View", "Run", "Kernel", "Settings", and "Help" menus. Below the top bar is a toolbar with icons for file operations and execution. The main area displays a terminal window with the following output:

```

llama --distance-metric mips
INFO:leann.embedding_server_manager:Server process started with PID: 167069
INFO:leann.embedding_server_manager:Embedding server is ready!
INFO:leann.api: Launching Server time: 1.5045411586761475 seconds
INFO:leann.embedding_server_manager:Reusing in-process server
INFO:leann.api: Generated embedding shape: (1, 768)
INFO:leann.api: Embedding time: 0.2468271255493164 seconds
INFO:leann_backend_hnsw.hnsw_backend: Search time in HNSWSearcher.search() backend: 0.375276565517578 seconds
INFO:leann.api: Search time in search() LEANN searcher: 0.37604594230651855 seconds
INFO:leann.api: Backend returned: labels=1 results
INFO:leann.api: Processing 1 passage IDs:
INFO:leann.api: ✓ [ 1] ID: '0' Score: 0.7527 Text: LEANN saves 97% storage compared to traditional vector databases.
INFO:leann.api: ✓ Final enriched results: 1 passages
INFO:leann.api: Search time: 2.1316916942596436 seconds
INFO:leann.chat:Sending request to Ollama and waiting for response...
INFO:leann.api: Ask time: 1.1547119617462158 seconds

💬 Answer: LEANN reduces the storage footprint by **97%** compared to traditional vector databases—meaning you only need about 3 % of the space that would otherwise be required.

```

The terminal output shows the execution of a llama.cpp command to generate an embedding and search for a passage. The results indicate that LEANN significantly reduces the storage footprint compared to traditional vector databases, saving 97% of the space. The final answer is: "LEANN reduces the storage footprint by **97%** compared to traditional vector databases—meaning you only need about 3 % of the space that would otherwise be required."

[illegible]

Let us now check out the LEANN CLI, we are again using the data we had

```
Ubuntu@0360-kci2-ty6k-prxm100037: ~/leann
Reading vector (dtype=<class 'numpy.float64'>, fmt='d')... Count=6, Bytes=48
[0.00s] Read assign_probab (6)
Reading vector (dtype=<class 'numpy.int32'>, fmt='i')... Count=7, Bytes=28
[0.29s] Read cum_nneighbor_per_level (7)
Reading vector (dtype=<class 'numpy.int32'>, fmt='i')... Count=2116, Bytes=8464
[0.58s] Read levels (2116)
[0.87s] Probing for compact storage flag...
[0.87s] Found compact flag: False
[0.87s] Compact flag is False, reading original format...
[0.87s] Probing for potential extra byte before non-compact offsets...
[0.87s] Found and consumed an unexpected 0x00 byte.
Reading vector (dtype=<class 'numpy.uint64'>, fmt='Q')... Count=2117, Bytes=16936
[0.87s] Read offsets (2117)
[1.17s] Attempting to read neighbors vector...
Reading vector (dtype=<class 'numpy.int32'>, fmt='i')... Count=137312, Bytes=549248
[1.17s] Read neighbors (137312)
[1.47s] Read scalar params (ep=1751, max_lvl=2)
[1.47s] Checking for storage data...
[1.47s] Found storage fourcc: 49467849.
[1.47s] Converting to CSR format...
[1.49s] Conversion loop finished.
[1.49s] Running validation checks...
Checking total valid neighbor count...
OK: Total valid neighbors = 44594
Checking final pointer indices...
OK: Final pointers match data size.
[1.49s] Deleting original neighbors and offsets arrays...
CSR Stats: |data|=44594, |level_ptr|=4291
```

```
Ubuntu@0360-kci2-ty6k-prxm100037: ~/leann
[0.58s] Read levels (2116)
[0.87s] Probing for compact storage flag...
[0.87s] Found compact flag: False
[0.87s] Compact flag is False, reading original format...
[0.87s] Probing for potential extra byte before non-compact offsets...
[0.87s] Found and consumed an unexpected 0x00 byte.
Reading vector (dtype=<class 'numpy.uint64'>, fmt='Q')... Count=2117, Bytes=16936
[0.87s] Read offsets (2117)
[1.17s] Attempting to read neighbors vector...
Reading vector (dtype=<class 'numpy.int32'>, fmt='i')... Count=137312, Bytes=549248
[1.17s] Read neighbors (137312)
[1.47s] Read scalar params (ep=1751, max_lvl=2)
[1.47s] Checking for storage data...
[1.47s] Found storage fourcc: 49467849.
[1.47s] Converting to CSR format...
[1.49s] Conversion loop finished.
[1.49s] Running validation checks...
Checking total valid neighbor count...
OK: Total valid neighbors = 44594
Checking final pointer indices...
OK: Final pointers match data size.
[1.49s] Deleting original neighbors and offsets arrays...
CSR Stats: |data|=44594, |level_ptr|=4291
[1.80s] Writing CSR HNSW graph data in FAISS-compatible order...
Pruning embeddings: Writing NULL storage marker.
[2.13s] Conversion complete.
Index built at /home/Ubuntu/leann/.leann/indexes/my-demo/documents.leann
(leann) Ubuntu@0360-kci2-ty6k-prxm100037: ~/leann$
```

```
Ubuntu@0360-kci2-ty6k-prxm100037: ~/leann
(leann) Ubuntu@0360-kci2-ty6k-prxm100037: ~/leann$ leann search my-demo "What techniques does LEANN use?"
[read_HNSW - CSR NL v4] Reading metadata & CSR indices (manual offset)...
[read_HNSW NL v4] Read levels vector, size: 2116
[read_HNSW NL v4] Reading Compact Storage format indices...
[read_HNSW NL v4] Read compact_level_ptr, size: 4291
[read_HNSW NL v4] Read compact_node_offsets, size: 2117
[read_HNSW NL v4] Read entry_point: 1751, max level: 2
[read_HNSW NL v4] Read storage fourcc: 0x6c6c756e
[read_HNSW NL v4 FIX] Detected FileIOReader. Neighbors size field offset: 59906
[read_HNSW NL v4] Reading neighbors data into memory.
[read_HNSW NL v4] Read neighbors data, size: 44594
[read_HNSW NL v4] Finished reading metadata and CSR indices.
INFO: Skipping external storage loading, since is_recompute is true.
```

We can now start searching the created index


```
Ubuntu@0360-kci2-ty6k-prmx100037: ~/leann
2. Score: 1.064
   Finally, we note that when a target recall level (e.g., 90%)
   is enforced, the downstream accuracy of our method aligns
   with that of other lossless ANN approaches, confirming that
   our system does not s...

3. Score: 1.052
   Hence, it can scale efficiently to large
   datasets, providing a simple yet effective mechanism to bal-
   ance graph size and search performance.
6
Evaluation
In our evaluation, we answer the following im...

4. Score: 1.048
   As a result, LEANN frequently fails
   to meet the minimum batch size required to saturate GPU
   throughput, leading to inefficient use of hardware resources
   at runtime.
   This leads to a key challenge: how ...

5. Score: 1.021
   Fig. 4 shows that LEANN consistently outperforms Edge-
   RAG, an IVF-based recomputation method, achieving signifi-
   cantly lower latency, ranging from 21.17x to 200.60x, across
   all the datasets and hard...

(leann) Ubuntu@0360-kci2-ty6k-prmx100037:~/leann$
```

```
Ubuntu@0360-kci2-ty6k-prmx100037: ~/leann
(leann) Ubuntu@0360-kci2-ty6k-prmx100037:~/leann$ leann list
LEANN Indexes
=====
🌟 Current Project
/home/Ubuntu/leann
-----
1. 📁 my-demo ✅
   📦 Size: 2.2 MB
2. 📄 leann ✅
   📦 Size: 0.0 MB
=====
📊 Total: 2 indexes across 1 projects

🔑 Quick start (current project):
leann search my-demo "your query"
leann ask my-demo --interactive
(leann) Ubuntu@0360-kci2-ty6k-prmx100037:~/leann$
```

We can also check the available LEANN embedded projects as above